

strength of each method and by combining their respective strengths within one single research design. The argument of incompatibility between two methods on philosophical ground is not valid because of the following reasons: (Bergman, 2008)

- i) The QL and QN methods represent large and heterogeneous families of methods which vary within their own family to such an extent that it is difficult to identify unique set of qualities or attributes that encompasses the characteristics of one family of methods, clearly distinct from the characteristics of the members of the other family. Most characteristics encompass either only a subgroup of members of the family or are also applicable to some members of other family. Hence there is a need to re-think the division of labour between QL and QN methods in order to better understand the possibilities and functions of methods to better justify and apply mixed method design.
- ii) The proposition of existence of one single reality under QN method is inconsistent with research applications. The emergent structures are based on co-construction between the researchers' selection and understanding of items in a questionnaire, their choice of analytic strategy and their interpretation of the statistical output on the one hand, and the way the respondents interpreted the survey questions with the given social, political, historical and economic context, on the other hand.

From a methodological perspective, it is not proper to declare one approach more or less valid, valuable or scientific. Instead how to understand and analyze data need to be based to a large extent on the consistency formed between how to understand data in conjunction with the specific research question, rationale, aim etc. only in connection with the specificities of the research goals it makes sense to delimit the nature of reality. Thus, the decision on whether the researcher deals with one single reality, a constructed reality, multiple realities, multiple constructed realities or a co-constructed reality is unrelated to whether patterns in the data are detected via statistical analysis or otherwise.

- iii) *A small vs large samples:* The sample size for many QN studies is often quite small and sample size for some QL studies can indeed be large.
- iv) Regarding hypothesis testing, many researchers engaged in QL research, pursue conjectures that are embedded either in their research questions, the kind and the way they collect data, the way they analyse the data and the way, they protect themselves from selective reportings of their findings. In many types of well-established statical analyses, the QN methods like cluster analysis, factor analysis, correspondence analysis, multi-dimensional scaling are used in exploring data structures.

In order to avoid misconceptions and mistakes while deciding the research design issues, the following points need to be kept in mind.

- a) We refrain ourselves from making specific argument for or against the QL and QN method for a specific research project. We need to focus our efforts more explicitly on embedding and justifying our selected methods according to our research questions, data needs, theoretical grounding and research design.

- b) Data collection methods (i.e. unstructured narrative interview, survey research based on closed-ended questions) and data analysis methods (qualitative content analysis, discourse analysis, quantitative content analysis) should be differentiated.
- c) We ought to be more aware of the actual inductive and deductive analytic phases of our research projects.

Check Your Progress 2

- 1) Which constituent of research study does guide the research design?

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- 2) State the functions of research design?

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- 3) What is the distinction between Mono Method and Mixed Method?

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6.6 THE RATIONALE FOR MIXED METHODS RESEARCH

Mixed methods research has gained tremendous popularity in the social, behavioural and related sciences in recent years. This increasing popularity of mixed methods research is reflected in terms of the claims by mixed method researchers, increase in the publications on this topic and the inclusion of mixed methods designs in the text books. The argument for combining qualitative and quantitative methods is put forwarded on the following grounds.

- i) The research methods associated with both quantitative and qualitative research have their own strengths and weaknesses. Combining them allows the researchers to offset their weakness to draw on the strengths of both.
- ii) Mixed methods designs enable the researcher to provide a comprehensive account of the area of the enquiry in which he or she is interested if both quantitative and qualitative methods are used. for example, causal explanation of any event or phenomena can be captured by quantitative methods whereas the reason explanation and pattern explanation or the explanation involving processes can better be captured by qualitative methods.

- iii) Quantitative and qualitative methods be used to answer different questions in a study. Further mixed methods can be utilized in order to gain complementary views about the same phenomenon or relationship. Research question for the two strands of the mixed study address related aspects of the same phenomenon.
- iv) Mixed methods design can be used to develop the research further. Questions for one strand emerge from the inferences of a previous one (sequential mixed methods), or one strand provides hypothesis to be tested in the next one.
- v) Mixed methods may also be used to assess the credibility of inferences obtained from one approach (strand). Exploratory and explanatory/confirmatory questions in such situation, may be useful.
- vi) Mixed methods are also used to obtain divergent pictures of the same phenomenon. These divergent findings can ideally be compared and contrasted.
- vii) Quantitative research provides an account of structures in social life but qualitative research provides sense of process.
- viii) Mixed methods research entails to generate hypothesis by using qualitative method and testing the hypothesis resorting quantitative method within a single project.
- ix) Mixed methods research enables to present diversity of views i.e. combining researchers' and participants' perspectives through quantitative and qualitative research respectively by way of uncovering relationships between variables through quantitative research and revealing meaning among research participants through qualitative research.

6.7 FORM OF MIXED METHODS RESEARCH DESIGNS

Connecting qualitative and quantitative methods through sequencing and prioritising, there can be four forms of mixed methods research designs.

- 1) **Preliminary qualitative inputs to core quantitative method (qual-QUANT):** With a view to use the strengths of qualitative methods to contribute to core quantitative methods, preliminary qualitative methods are used for collecting and interpreting the quantitative data. Preliminary qualitative methods can also be used as an input to generate hypotheses, develop content for questionnaires etc.
- 2) **Supplementary qualitative method as a follow up to core quantitative method (QUANT-qual):** Supplementary qualitative method as a follow up to core quantitative method can be used to extend what is learned from quantitative study. It can demonstrate bases for results, examine reasons for failed hypotheses, explore the theoretical significance of outliers and so on.
- 3) **Preliminary quantitative inputs to core qualitative methods design (quan-QUAL):** The study undertaken by using core qualitative method design gets inputs by way of using preliminary quantitative method for collecting and interpreting the qualitative data. Preliminary quantitative

method can also locate major differences between sub-groups, guide purposive sampling, establish results to pursue and so on.

- 4) **Supplementary quantitative method to core qualitative method design (QUAL-quan):** Supplementary qualitative method as a follow up to core qualitative method can be used to develop measures of key concepts and relationships across different samples.

6.8 CASE STUDIES OF MIXED METHODS RESEARCH DESIGN

- i) Studies where the quantitative component is dominant in terms of the framework of positivism and post positivism paradigm and inductive and deductive logic of enquiry guides the study, quantitative approach is more dominant and the researchers specialize in quantitative work. In such studies, qualitative data and their manipulation are shaped by the nature of research questions raised in the quantitative part of the project. The qualitative data are treated and transformed in effect into quantitative variables with the purpose of fleshing out the explanations required for the quantitative results. In such cases quantitative component follows qualitative component mixed methods. An illustration of a Research Study which has used Mixed Method pre dominated by the Quantitative Method is given below in the box:

CASE 1: Mixed Method Research Design: Conducted the study within the framework of post positivist paradigm using inductive logic of inquiry and analyzing the qualitative data by quantitative technique of data analysis

**Study: ‘Decent Work: Concept, Measurement and Status in India’:
Ph.D thesis by Ms. Nausheen Nizami Ph.D Research Scholar,
(Economics) IGNOU, N. Delhi**

The study was conducted within the framework of positivist paradigm using inductive logic of enquiry with the objectives to (i) examine the applicability of indicators of ‘decent work in India (ii) develop indicators and tools to measure ‘decent work (iii) evaluate the status of decent work in IT companies and to identify the socio-economic attributes influencing status of decent work, and (iv) know the employer’s perspective in provision of decent work’.

Methodology: Decent work sums up the aspirations of the people in their professional life and is a revolutionary agenda of International Labour Office. Decent work is any such work which ensures provision of fair and free employment to all men and women of economically productive age-group in conditions of fairness, equity, security and dignity. It is a multidimensional concept that applies to both formal and informal sectors of an economy. The study is based on primary data collected through well designed questionnaire and telephonic interviews. Measurement of decent work was undertaken using both quantitative and qualitative methods of data analysis. A characteristic feature of this study was to transform the qualitative data in a manner so as to be able to use quantitative techniques of data analysis.

Quantitative Techniques to analyse qualitative data

a) Composite Index construction for Decent Work

With a view to measure the extent and range of decent work provision, decent work indices were constructed. These indices have been made for all the indicators of decent work used in this study by coding, normalising and aggregating the responses as the data was qualitative in nature. Thereafter, the standard formula developed by United Nation's Development Programme (UNDP) for the construction of HDI and other indices was applied.

b) Factor analysis of decent work indicators

The technique of factor-analysis was adopted to short-list the most relevant and reliable indicators of decent work. The principal component method was used to outline components (i.e. indicators in this study) explaining the maximum variance in the dependent variable.

c) Phi ϕ Coefficient of Correlation Analysis

Since the variables were dichotomous in nature, correlation analysis was attempted to test whether there was statistically significant associations between the responses on different decent work indicators. Partial correlation analysis examined the relationship between few socio-demographic variables and composite Decent work Index keeping the effect of other variables constant.

d) Chi-square test for independence

Chi-square test was conducted to test the independence of the results between the two scales of Decent work index (For example, 0 and 0-1) and results of each decent work indicator for those two scales.

Thus an efficient mix of quantitative and qualitative methods were used to deal with the key issues related to decent work in this doctoral thesis.

Major Results:

- 1) Composite decent work index reflects deficit in the status of decent work for a vast majority (94%) of IT employees.
- 2) Adequacy of earnings and productive employment are directly associated with decent work which is in consonance with the features of search and matching theory of employment.
- 3) Longer working hours have a significant impact on the physical and emotional well-being of an employee. A vast majority of IT employees (95%) were found to be working for longer hours.
- 4) Work-life balance and adequate earning and productive employment were found to be correlated. Majority of the employees in IT sector were found to be earning adequate earnings but their work-life balance was disrupted owing to several reasons including longer working hours. This implies that higher earnings of the employees in the IT industry are necessary

but not a sufficient condition for the employees. Their higher earnings are not resulting into their well-being (mental) as reflected in their deteriorating health status and imbalance between professional and personal lives.

- 5) Social security mechanisms, social dialogue and fair treatment at work places are major aspects of decent work owing to ensure sustainability of economic well-being of the employees.
 - 6) Decent work reflects decent work place but a decent work place may not always translate into decent work because of the gap of the nature of work and work amenities and environment. This implies that deficit in decent work to the employees is correlated with deficit in decent work places.
 - 7) Age and social class are the prominent attributes in influencing decent work status.
 - 8) Despite the presence of premier educational institutions like IITs, RECs, etc. demand-supply gap in the available manpower for IT industry is the major concern of the employers pushing them to face the problem of adverse selection. A direct implication of this situation is the low employment elasticity of IT industry with respect to its growth and increasing contribution to GDP.
- ii) Studies where qualitative component had priority in terms of the framework of interpretative paradigm and abductive logic of enquiry. In such situations, the researchers identify themselves primarily as qualitative researchers. In such studies, the tools used for data/information collection is mixed in nature. The modalities of the different types of data are maintained and are not treated as commensurate. Integration of data takes place as a part of interpretative process referred to as analytic or interpretative integration. In such studies quantitative techniques can be used to analyse the perceptions of the people which may be qualitative in nature. An illustration of the research study using mix of qualitative and quantitative methods in the manner discussed here in is given below:

CASE 2: Mixed Methods Research Design: Conducted the study within the framework of Qualitative Research Approach using Quantitative Methods for data analysis

Study: Gender specific impacts of climate variation in Agriculture:

By Mamta Mehar, Ph.D Research Scholar, IGNOU, N. Delhi

The Main Objective of the study was to understand the gender differences in agriculture and analyse the linkages between gender and climate variation using mixed method approach. The research questions addressed in this study were: (i) Does there exist difference in gender perception on climate variability? (ii) Whether male and female farmers follow similar coping strategies to mitigate risk and shocks in agriculture? The study focused on to understand how gender and climate variation in agriculture are interrelated with the support of empirical evidences.

Methodology

This study was conducted within the framework of qualitative approach anchored in interpretative paradigm using quantitative methods – questionnaire as a tool and survey as a method to collect the data and multivariate probit model to find out probability in deciding adaption strategy by male and female to climate variability.

This research study used case study method within a socio-cultural framework. Case study methodology derives much of its rationale and methods from ethnography characterized by “an instance in action” in a “real life context when the boundaries between the phenomenon and context are not clear”. Keeping in view, the logic of enquiry, the research questions were analysed using both qualitative and quantitative methods. The qualitative analysis was done within the framework of “Interpretative paradigm of qualitative research”. Since the research objective was to understand human behaviour within the surrounding context of climate change, where it was difficult to isolate the phenomena of climate variation and perception of males and females towards it, conducting of the present study under interpretive paradigm was appropriate. The associated research strategy used in this study was abduction. Abduction is the logic used to construct descriptions and explanations that are grounded in the everyday activities as well as in the language and meanings used by social actors. The social actors in the present proposal are men and women farmers, and their activities to be analysed are specific to decision making in climate change situation. The interpretative paradigm using abduction strategy was done via literature review as well as synthesis from findings of focus group discussion during the survey. In the wake of theoretical understanding, it was tried to quantify the impact using appropriate empirical analysis and using data from a baseline survey conducted by CCAFS in 2013 of 641 farm households in twelve villages of Vaishali districts in Bihar, India; it was attempted to quantify the gender differentiated impact.

Results

The linkage between gender and climate variation in agriculture was more influenced from subjective experiences of individuals or society. The movement of understanding “is constantly from the whole to the part and back to the whole” (Gadamer, 1976). According to Gadamer, it is a circular relationship. Since the study aims to understand the perception of human in context of climate variation, it cannot be explained entirely by any scientific paradigm where hypothesis are pre-structured and to some extent results are known. The interpretative paradigm points out that positivism, in its first attempt to model the social after the natural sciences, fails to see that unlike nature, social reality exists only insofar as lay members create that reality in meaningful interaction (Fuchs S., 1992: 198). But the point of interpretive paradigm is that we must first understand social worlds from within, before we develop scientific models and explanations (Fuchs, S. 1992: 205). According to Willis (1995) interpretivists are anti-foundationalists, who believe there is no single correct route or particular method to knowledge. Walsham (1995) argues that in the interpretive tradition there are no ‘correct’ or ‘incorrect’ theories. Instead, they should be judged according to how ‘interesting’ they are to the researcher as well as those involved in the same areas.

The linkages between gender and climate variation in context of agriculture are bi-directional. Changes in climate event effect agriculture through reduction in yield, increase in occurrence of pests or crop disease, changes in time of activities for example, changes in winter time could result in change in harvesting or sowing days of respective crops. All this affects farmers' livelihood and work activity in different ways. For example, women (and children) being the ones traditionally charged with water collection roles, have to travel longer distances after drought. Evidences suggest decrease in yield and thereby total availability of food consumption in a household basket usually has been seen as complemented by reduction in meals of female. This exposes them to health issues. Male farmers are usually seen to migrate in search of nonfarm opportunities outside their village domain, putting all burden or responsibilities on women. Moreover mostly in rural areas women are less likely to get education as compare to male, they are less allowed to mobile and do more household related activities, and they have less control over the family assets.

The other direction of gender differentiated impacts in context of climate events in agriculture can be viewed through understanding their coping strategies. The adaption of particular strategies are often influenced by the priorities and options available for women and men for coping these strategies. Constraints on women's mobility, and behavioural restrictions hinders their ability to make decision on various matters and thus they have less choices of coping strategies available. Additionally inequitable access to assets and resources such as land, knowledge, technology, limited power in decision-making, education, health care and food have been identified as determinant factors behind adaptation of different coping strategies of male versus female. The results of quantitative analysis also supports this argument.

The females are mostly illiterate (71 per cent) and have little access to resources. Only 9 per cent of farm household have reported to provide land entitlement to at least one female member. Additionally the gender power dynamics is seen to favour only one gender i.e. male. The decision making role in different household related expenditures as well as in agriculture activities is dominated by male. Considering the perception of male vis-à-vis female farmer about effects of climate change has shown a similar pattern. More than 90 per cent of farmers are aware about climate variation and feels that weather conditions adversely affect their agriculture activities. However again, the decisions to cope with the risk due to climate have suggested different behaviour, priorities by male and female. Survey results suggest that almost 60 per cent of surveyed farmers have adapted at least one coping strategy and in total adoption of more than 30 coping strategies are reported. The results of multivariate probit model suggest that male farmers have higher probability in deciding adaptations through additional jobs as well as specific agriculture strategies such as crop rotation and use of hybrid seed in farming practices.

- iii) Studies presenting diversity of views i.e. combining researcher's and participants' perspectives through quantitative and qualitative research. Researchers uncover relationship between variables through quantitative methods and reveal meaning among research participants through qualitative methods. Thus a comprehensive account of the area of the enquiry is presented. An illustration of the study using this form of mixed method research is given below:

CASE 3: Mixed Method Research –used Quantitative Methods for data analysis and Qualitative Methods for revealing and understanding participants' perspective on quality of education

Study: 'Measuring Quality of Education and Its Determinants: Indian Context': Ph.D thesis submitted by Ms. Charu Jain, Ph.D Research Scholar, (Economics) IGNOU, New Delhi

The study was carried out with the objectives to (i) examine the linkages of secondary education with various socio-economic outcomes at all India level and state level, (ii) identify various determinants (including students' family background characteristics, and schools' characteristics) affecting students' performances and teaching efficiencies in senior secondary schools in Delhi. To fulfill these objectives, both quantitative and qualitative research approaches were applied.

Quantitative Method: The quantitative approach was used to test the hypothesis generated. Two set of questionnaires one related to the teachers and the other to students were got filled up through in-depth face to face interviews with the respondents. The student questionnaire was translated in Hindi language to enhance the understanding level of students in Hindi medium government schools in Delhi. Within translation procedure, the student questionnaire was first translated in Hindi and thereafter re-translated in English to re-check the accuracy and meaning of questions. The teachers' questionnaire was provided in English language only. The information thus retrieved from these questionnaires was used to measure the performance of students and teacher analyzing thereby the overall quality of education at lower secondary and senior secondary levels of education. The information collected through quantitative approach was statistically analyzed and tested using various statistical softwares like SPSS and STATA?

Qualitative Method: In the qualitative approach, direct observation method and key informer interviews were used to supplement the information/data collected through quantitative method by way of two sets of structured questionnaires. Qualitative research method enabled the researcher to approach the research topic from the perspective of teachers and students. Further, the qualitative techniques were used to examine social processes which otherwise could not have been captured by traditional quantitative measures.

The direct observation technique enabled researcher to learn about the behaviour of the people under study: students and teachers in the natural setting i.e. schools through observing. It gave the researcher the first hand information on the quality aspects of school and classrooms and working conditions for teachers, which were quite useful in doing the analysis.

The field notes, which were taken down during the period of fieldwork, were instrumental during analysis and interpretation of the results, as they helped in recalling the incidents that took place at the time of data collection. *Face-to-face interviews* with the key informants were also conducted within qualitative approach. Key informant interviews were qualitative informal interviews with people who knew what was going on in the community. Our purpose of conducting key informant interviews was to collect information from people who had firsthand knowledge about the overall functioning of school. The key informers in the sample schools were selected using purposive sampling technique and in this case the key informants were either senior teacher in school or senior administrative person in schools. Interviewing key informants enabled the researcher to look at the underlying issues and problems with varying perspectives. These interviews were conducted using an informal approach. Initially the questions were drafted in a form of well designed open ended short questionnaire for conducting face to face interviews with the respondents. One respondent or a group of 2-3 people from each school were selected purposively for gathering this information, depending upon how well they were informed about the school.

Apart from these informal interviews, the school guards/receptionists were also contacted informally to gather the information on schools. The information thus collected through qualitative approach was first entered in Excel in a pre-designed format and then coded so that it can be further analysed statistically. Moreover, the responses to few open ended questions were reviewed and inferences were drawn which were quite useful in providing suggestions and recommendation for this study.

Main Results

- 1) Secondary education increases income level of individuals to a significant extent. One unit increase in Gross Enrollment Ratio (GER) at secondary level leads to increase in the personal disposable income by 0.65 units.
- 2) The educational background of households determine their occupational pattern. The households with at least secondary education are either salaried or business class, while those with elementary level of education are more concentrated in agricultural/wage earning activities. Higher the level of education among females, greater the proportion of female work participation rate.
- 3) Secondary education bears highest impact on improving health indicators. Secondary education has high potential for bringing demographic transitions in terms of decline in maternal mortality rates, infant mortality rates, fertility rates, death rates etc.
- 4) Secondary education attainment brings behavioral changes in individuals in terms of higher savings and diversion in the expenditure pattern.
- 5) School resources, teachers and teaching quality, student's family background, mass media exposure and self motivational factors of students determine their outcome to significant extent.
- 6) Within school characteristics, cleanliness in school, well qualified teachers their friendly and supportive attitude with students and quality of school infrastructure influence students performance positively. Lack of student motivation, large class size, lack of school resources and facilities, lack of adequate teaching material affect the teaching abilities adversely.

Check Your Progress 3

- 1) Give two situations where Mixed Method research is superior to Mono Method?

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- 2) Give an example of Mixed Method where qualitative method predominates quantitative method?

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- 3) State the different forms of Mixed Methods?

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6.9 LET US SUM UP

Traditionally two distinct research approaches – Quantitative (QN) and Qualitative (QL) – have been followed in undertaking research in social sciences. The former is associated with positivism and post positivism paradigm and the latter principally with interpretative paradigm and critical theory paradigm. Depending upon the nature and field of specialization research studies may be of various types. Important among these are – theoretical and applied, descriptive and explanatory, conceptual and empirical, exploratory and experimental etc.

Research design is a logical structure of enquiry specifying the types of evidences needed to answer the research questions. Research methods are made of tools for data collection, type of data, sampling design, sample size, techniques for data analysis etc. There is nothing intransigent about any research design that requires a particular method of data collection. Research Methods can broadly be of two types – Mono Methods and Mixed Methods. Within mono methods, it can be either quantitative or qualitative. The combination of at least one qualitative and at least one quantitative component in a single research project is known as mixed methods research. The mixed methods have gained popularity in carrying out research in social and behavioral sciences because combining them allows the researchers (i) to offset their (QN and QL methods) weaknesses, (ii) provide comprehensive account of the area of the enquiry, (iii) to answer different types of questions. Studies undertaken under mixed methods can be of four types – (i) preliminary quantitative methods